THE UNITED REPUBLIC OF TANZANIA

FORM FOUR PRE-MOCK EXAMINATION

FOOD AND HUMAN NUTRITION

MARKING SCHEME

SECTION A (16 MARKS)

1.

i	ii	iii	iv	v	vi	vii	viii	ix	x	
C	A	C	B	B	A	B	B	D	B	

1 mark @ = 10 marks

2.

List A	i	ii	iii	iv	v	vi
List B	D	H	G	C	L	A

1 mark @ = 6 marks.

SECTION B 54 marks

3. (a) Whole meal cereals nutritionally preferred to refined one because of the following reasons

(i) High in nutrients because they retain all parts of the grain, the bran, germ and endosperm

(ii) High in fibre which helps with digestion, prevents constipation and supports a healthy gut.

(iii) Better for blood sugar control

(iv) Supports heart health.

(v) Provides protein and healthy fats.

1 mark @ = 7 marks.

3(b) Advice for a person who does not consume animal protein.

i) To include nuts and seeds / plant protein source in the diet.

ii) Ensure adequate iron intake

iii) Include Omega – 3 fatty acids

1 mark @ = 2 marks.

(c) Importance of dietary fibre in human health.

i) Aids digestion

ii) Lower cholesterol in the gut.

iii) Helps in weight management

iv) Prevents digestive disorders

v) Supports guts health.

vi) Speeds up the passage of food through the digestive tract, hence increasing faecal bulk.

1 mark @ = 2 marks

Total = 06 marks

4 (a) Malnutrition is a condition that happens when the body does not get right amount of nutrients for long time. 2 marks

b) Malnutrition diseases with their deficiency nutrients

i) Kwashiorkor — Protein

ii) Marasmus — Severe lack of calories.

iii) Anemia — Iron.

iv) Scurvy — Vitamin C

v) Rickets (in children) / osteomalacia (in adults) — Vitamin D.

vi) Nightblindness – Vitamin A.

vii) Goiter — Iodine

viii) Beriberi — Vitamin B_1

ix) Pellagra — Vitamin B_3 (Niacin)

02 1/2 mark @ = 2 marks

4 (x) The name of the given nutrients

(i) Vitamin A — Retinol

(ii) Vitamin B_1 — Thiamine

(iii) Vitamin B_2 — Riboflavin

(iv) Vitamin C — ~~Scurvy~~ Ascorbic acid

1 mark @ = 4 marks.

Total marks ~~10~~ (08)

5(a) Nutrient supplements are popular for several important reasons

(i) Busy life style

Many people have fast-paced lives and rely on processed or convenience food, which lack some important nutrients.

(ii) Poor eating habits

Not everyone eats a balanced diet with enough fruits, vegetables, proteins and whole grains.

(iii) Health awareness.

People are more aware of health and wellness today.

(iv) Medical needs

Some groups truly needs supplements such as pregnant women (folic acid, iron) Elderly people (vitamin D, B_{12}) etc. ½ mark point, ½ mark explanation

b) Effects of taking large amount of supplements

(i) Can cause birth defects (in pregnancy).

(ii) Liver damage

(iii) Kidney stones (in very high amounts.

(iv) Allergy. 1 mark @ = 2 marks

~~½ point @ and ½ explanation @~~

03. = 04 marks

05 (c) The following nutrients are needed because of the following

(i) calcium — For strong bones and teeth.
— For muscle function.
— Blood clotting — necessary for blood clotting when you get a cut or injury.
— Never transmission

1 mark @ = 02

(ii) Iron — Make Haemoglobin.
— It prevent anemia
— Support energy production
— Boosts immunity.

(d) Kinds of foods that provides these nutrients.

(i) Calcium — Milk, cheese, yogurt, Sardines, beans green leafy vegetables

½ ½ mark @ = 1 mark.

(ii) Iron — Red meat, liver, beans, fish, Spinach fortified cereals.

½ mark @ = 1

Totall 10 marks.

06. (a) First course — consomée Soup

(b) Second course — Vegetable rice
— Beef stew
— Sautee chinese

Third course — Jam pudding

1 mark @ course = 3 marks

6. (b) Different between meal and dish

A meal is a food eaten at a particular time e.g A dinner meal might have rice, curry, salad and dessert.

while

A dish is a specific prepared food item cooked at a one pot for example Fried rice, chicken curry, beef and banana stew.

1½ @ = 3 marks

(c) Points to consider when making Stock

(i) Use fresh Ingredients
(ii) Choose the right bones ~~and~~ or vegetables
(iii) Keep cold water (use cold water).
(iv) Simmer, Don't boil
(v) Skim fat and impurities.
(vi) Add Herbs and Seasoning carefully
(vii) Strain stock
(viii) Cool and store properly.

½ mark @ = 4 marks

Total 10 marks

7. (a) How to prevent food contamination

(i) By practising personal hygiene
(ii) To clean food and utensils
(iii) By proper storage of food. e.g keep raw food separate from ready-to-eat foods.
(iv) To cook food properly in order to kill Harmfull microorganisms
(v) To avoid cross contamination e.g use separate chopping boards for meat and vegetables.

1 mark @ = 5 marks.

7. (b) Ways in which leads food decaying.

(i) Microbial action – Bacterea, molds and yeast grow on food.

(ii) Enzymatic action – This leads to changes in color, taste, and texture.

(iii) chemical reaction – Some foods spoils due to oxidation or other chemical reactions.

1 mark @ = 3 marks

(c) Food can become poisonous to human after being consumed due to contamination or harmful substances.

2 marks

Total of 10 marks.

8. (a) Mechanical methods of introducing air into a mixture.

(i) Beating – rapidly mixing ingredients with a spoon, whisk, or electric mixer to incorporate air.

(ii) whisking – using a whisk or electric beater to trap air bubbles in a liquid.

(iii) Folding – Gently mixing a lighter mixture into a heavier one without losing the air.

(iv) Creaming – Beating fat with sugar to trap tiny air bubbles.

(v) Sieving or sifting – Passing flour or powdered ingredients through the sieve to lighten them and tiny air pockets.

(vi) Kneading (for yeast dough) Working dough to incorparate air and develop gluten, which traps gas.

½ points and ½ explanation

= 04 marks

8 (b) Points to remember when using raising agents
(i) Measure accurately
(ii) Mix evenly.
(iii) Use fresh raising agents
(iv) Pay attention to liquids
(v) Bake or cook immediately
(vi) Know the type of raising agents
1 mark @ = 3 marks

(c) What happen if too little raising agents used.
(i) Heavy texture
(ii) Poor volume
(iii) Tough or chewy food.
(iv) Uneven cooking
(v) Reduced appearance appeal
1 mark @ = 3 marks.
Total 10 marks

SECTION C (30 marks)

9. (a) Basic ingredients for making pastry.
(i) Fat (butter)
(ii) Wheat flour.
(iii) Liquid
(iv) Salt and Sugar
1/2 mark @ = 2 marks

(b) Types of pastry
(i) Short crust pastry
(ii) Flaky pastry
(iii) Rough puff pastry
(iv) Puff pastry
(v) Choux pastry
(vi) Suet pastry
2 marks @ = 8 marks.

9. (c) Rules for making pastry.

(i) Keep all ingredients and ~~ingredients~~ utensils as cool as possible except for the pastry which needs a warm condition.

(ii) Use enough liquid when making pastry dough and during binding the pastry.

(iii) Do not re-roll the pastry as it will crack during cooking

(iv) Glaze pastry dish before baking to give a glossy surface.

(v) Use little flour as possible for rolling out

1 marks @ = 5 marks.

Total 15 marks.

10. IRON

Introduction 2 marks.

sources of iron: Animal sources: Liver, Lean meat, egg yolk, poultry, fish.

Plant sources; Dark green vegetables, pulses, millet, sorghum, cocoa, and peanuts; Iron fortified foods.

Functions: Synthesis of Haemoglobin
- Assists in brain formation of foetus
- Gives blood its red colour.

Effects of its deficiency in the body.
- Anaemia
- Fatigue
- Dizziness/headache
- Pale skin/colour inside of the lower eyelids
- shortness of breath.

each point 12 marks.

conclusion: 01 mark

Total 15 marks.

11. Causes of malnutrition ~~and~~ marks

Introduction 01 mark.

causes:
- Inadequate dietary intake.
- shortage of food in the family
- Poor maternal and child care.
- Diseases.
- Poor hygiene and Sanitation
- Poor Health services
- Inadequate nutrition education
- Large family size.
- Poor economic status of the family.

1 mark @ = 9 marks

Ways to managing malnutrition.
- Exclusive breast feeding
- Food fortification
- Micronutrient supplementation
- Home gardening and ~~Live~~ Stock keeping
- ~~Tre~~atment and prevention of diseases
- Safe water supply, sanitation and hygiene.

1 mark @ = 4 marks.

Conclusion: 01 mark.

Total 15 marks